Informational Power Through Pivotality: How Consensus Can Benefit a Hegemon

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Abstract

When can consensus benefit a powerful state even without granting it proposal power? I study a two-round bargaining game in which only weak states propose, the hegemon privately knows its outside payoff, and a no vote by the hegemon is an immediate and irreversible opt-out. Under unanimity, weak proposers must satisfy the hegemon's private participation constraint. In the regular interior domain, a unanimity PBE exists only under derived incentive and belief conditions; whenever it exists, every regular on-path equilibrium pools at the high outside payoff. Under majority rule, equilibrium is generally set-valued: weak coalitions can exclude the hegemon, but larger groups can also sustain separating current passage. Institutional comparisons are therefore conditional on the common PBE-existence domain. On that domain, weak-state formation under unanimity is nested in formation under majority, while unanimity weakly raises the hegemon's payoff whenever both institutions form, with equality under majority pooling. Consensus can thus create informational power through pivotality even when agenda power is deliberately assigned to weaker states.

1 Introduction

When can consensus benefit a powerful state? A control-based intuition says that majority rule should be more attractive to powerful actors because a winning coalition can act without unanimous consent. Consensus seems to move power in the opposite direction by giving every participant a veto.

This paper identifies a different source of power. Consensus can favor a privately informed hegemon because it makes that actor's approval an informational constraint on weaker states. Weak states know the institutional package they can offer but not the hegemon's privately known outside payoff. When the hegemon is pivotal, the weak proposer must decide whether and how to insure agreement against that hidden participation constraint.

The model separates outside-option power, pivotality, and agenda power. The hegemon H is never recognized as proposer: $\pi_H = 0$ in both bargaining rounds. Only weak states propose. A proposal is a relative institutional package y, such as a quota share, production flexibility, an exception, or an enforcement advantage. It is feasible in every state and reduces the unit surplus left to the weak coalition one-for-one. If H votes no in Round 1, it opts out immediately and irreversibly, receives o_θ without discounting, and cannot re-enter. If H votes yes but simultaneous weak votes make the ballot fail, H remains active and the game can continue.

The clean timing changes the equilibrium architecture. First, terminal unanimity retains the familiar screening choice between a low-only package and pooling. Second, in the regular interior domain, Round-1 unanimity does not admit a global pooling/low-only/delay reduction. A global PBE exists only if $\beta o_1 \geq o_0$ and a strict security-value condition holds; whenever it exists, every regular on-path equilibrium pools at $y = o_1$. Third, majority is not a unique no-screening benchmark. Its exact security value is the maximum of exclusion, low-only, and pooling guarantees, and its equilibrium correspondence depends on group size. Finally, institutional comparisons are made only on the common PBE-existence domain. There, weak-state entry under unanimity is nested in entry under majority, and unanimity weakly benefits H whenever both institutions form.

These are conditional results, not claims that unanimity always dominates majority or that the model characterizes all PBEs under unrestricted beliefs. The analysis uses perfect Bayesian equilibrium together with a weak-vote-passive assessment: an uninformed weak voter's unexpected ballot is not treated as direct evidence about H's type. The assessment does not impose sequential-equilibrium consistency across globally off-path information sets.

2 Consensus, Information, and Institutional Design

The argument contributes first to rational-design accounts of international institutions. Existing work emphasizes how institutional rules help states manage uncertainty, commitment, and distributional conflict (Koremenos, Lipson, and Snidal 2001). Consensus is often interpreted as a device for protecting participants from unfavorable outcomes or encouraging broad participation (Gould 2016). The mechanism here reverses the usual direction of the informational benefit: uncertainty can make consensus valuable to the privately informed pivotal actor.

Second, the paper qualifies self-binding and informal-power accounts. Self-binding theories explain why powerful states may accept constraints to make cooperation credible (Keohane 1984; Ikenberry 2001). Informal-power accounts show how powerful states can dominate outcomes even under formally egalitarian rules (Steinberg 2002; Stone 2011; Gruber 2000). Here the hegemon neither proposes nor needs to control drafting. Its leverage comes from a private outside payoff that becomes politically productive when its approval is necessary.

Third, the model builds on bargaining theories in which recognition and voting rules determine distribution (Baron and Ferejohn 1989; Kalandrakis 2006; Eraslan and Evdokimov 2019). Setting $\pi_H = 0$ removes formal proposal power from the hegemon and isolates pivotality. The informational problem is related to veto bargaining (Bardhi and Guo 2018; Kim, Kim, and Van Weelden 2025), but there is no information designer or commitment to an experiment. Information matters because an institutional quota determines whether a privately informed vote is necessary for implementation.

Two recent papers provide close but distinct comparisons. Piazolo and Vanberg (2025) study a setting in which privately informed responders can signal through rejection and bargaining may break down. Glynia, Thum, and Xefteris (2026) analyze a one-shot ultimatum environment with a fixed proposer and a trade-off between insuring unanimity and gambling on a majority coalition.

The present paper asks whether consensus can benefit a powerful non-proposer. Its distinctive features are weak-state agenda control, an immediate and irreversible opt-out by the hegemon, and an explicitly conditional comparison over equilibrium correspondences.

3 A Motivating Terminal Example

Consider a terminal bargaining round with one hegemon and two weak states. The weak coalition has a unit surplus. A higher package y benefits H and reduces the weak residual one-for-one. The low type has outside payoff o_0, the high type has outside payoff o_1, and $0 < o_0 < o_1 < 1$. Because implementation is certain if H votes yes, type θ accepts exactly when $y \geq o_\theta$.

Under unanimity, a weak proposer compares a low-only package $y = o_0$, worth $(1 - \mu)(1 - o_0)$, with a pooling package $y = o_1$, worth $1 - o_1$. It pools when

$$\mu > \frac{o_1 - o_0}{1 - o_0}.$$

Pooling gives the low type an informational rent $o_1 - o_0$.

This terminal example is deliberately limited. In Round 1, ballots are simultaneous, a no vote by H is an immediate opt-out, and a yes vote followed by weak-caused failure leads to a distinct continuation. The direct cutoff $y = o_\theta$ therefore applies only when current implementation after H-yes is certain. The full model derives the expected voting incentive constraint rather than treating the terminal cutoff as a global rule.

4 The Model

Definition 1 (Clean immediate-opt-out game)**.** *There are $N \geq 3$ states: one hegemon H and $m = N - 1$ weak states. Nature draws a persistent type $\theta \in \{0, 1\}$. Only H observes its type; weak states share prior $\mu = \Pr(\theta = 1)$. The weak coalition has a fixed institutional surplus normalized to one, and weak outside payoffs are zero. The primitive domain is*

$$0 \leq o_0 < o_1 \leq \bar{y} \leq 1, \qquad \beta \in (0, 1]. \tag{1}$$

If a current agreement includes H, its payoff is y; hence $b_0 = b_1 = 0$. The outside payoff o_θ is external to the institutional pie.

The game has two rounds. In every reached round, a new weak proposer is drawn uniformly and independently; $\pi_H = 0$ in both rounds. If weak state i proposes while H is active, its contract is

$$s_i = (y, (x_j)_{j \neq i}), \qquad y \in [0, \bar{y}], \quad x_j \geq 0.$$

If H participates, the proposer receives

$$x_i^H = 1 - y - \sum_{j \neq i} x_j \geq 0. \tag{2}$$

If majority passage excludes H, the conditional component y is reabsorbed and the proposer receives

$$x_i^{WO} = 1 - \sum_{j \neq i} x_j = x_i^H + y. \tag{3}$$

Thus H's outside payoff is never charged to weak surplus, and H never receives both y and o_θ in the same history.

The proposer is counted as voting yes. Every other active player observes the proposal and votes simultaneously; individual votes become public only after the ballot closes. Unanimity requires N yes votes. Majority requires

$$q = \left\lfloor \frac{N}{2} \right\rfloor + 1. \tag{4}$$

The protocol is identical across rules except for the quota and is not a sequential roll call.

A no vote by type θ of H is an immediate and irreversible opt-out. In Round 1 it pays o_θ immediately and without discounting. No current or future agreement can then include H. Under unanimity the branch ends because the original quota N is impossible. Under majority, a current weak-only agreement passes if the weak yes votes already reach q; otherwise the game proceeds to a weak-only Round 2 with the original quota q. After opt-out, every proposal fixes $y = 0$ and uses the weak-only budget in (3)*. If H votes yes and the Round-1 ballot fails only because weak votes do not reach the quota, H remains active and the game proceeds to Round 2.*

Round 2 is terminal, and all Round-2 payoffs are multiplied by β in Round-1 units. If H is still active and a Round-2 ballot fails, H receives o_θ at the Round-2 date regardless of whether it voted yes or no, while weak states receive zero. If H opted out in Round 1, it receives no additional Round-2 flow.

Before Round-1 recognition, weak states make one collective, all-or-nothing formation decision and pay an external sunk cost χ each if the institution forms. For rule R, define

$$V_W^R(\mu) = \frac{1}{m}\mathbb{E}\left[\sum_{j=1}^{m} u_j^R \mid \textit{formation}\right]. \tag{5}$$

Formation occurs iff $V_W^R(\mu) \geq \chi$, with formation at equality. If the institution does not form, every weak state receives zero and type θ of H receives o_θ.

The immediate-opt-out timing is summarized in Figure 1. The distinction between an H-no and an H-yes followed by weak rejection is the core protocol distinction.

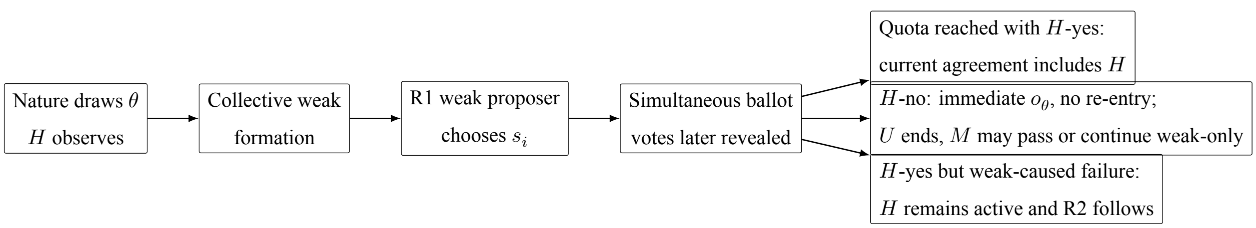

Figure 1: Timing of the clean immediate-opt-out baseline. Ballots are simultaneous. A no vote by H is an immediate and irreversible opt-out; a yes vote followed by weak-caused failure leaves H active for Round 2.

Definition 2 (Weak-vote-passive assessment)**.** *The primitives, type distribution, recognition rule, contract space, ballot protocol, quotas, implementation rules, payoff consequences, and tie-breaks are common knowledge. The solution concept is PBE. An assessment specifies the collective formation strategy; every weak proposer's proposal strategy at each public history where it is recognized; every active non-proposer's behavioral voting strategy; H's type-contingent behavioral voting strategy; and beliefs at every information set where θ remains payoff-relevant. Strategies are sequentially rational given those beliefs, and beliefs obey Bayes' rule wherever possible.*

A unilateral vote deviation by an uninformed weak state is not treated as direct evidence about θ; beliefs instead follow the public proposal and H's simultaneous type-contingent strategy and revealed vote. A separating vote by H can update beliefs when H remains active.

Every zero-probability continuation in which H remains active must state its off-path posterior. Globally off-path ballot beliefs and later zero-probability continuation beliefs are components of weak PBE and need not be linked by Bayes' rule. The assessment is not a refinement, D1, the Intuitive Criterion, a sequential-equilibrium restriction, or a characterization of all PBEs.

Payoffs from a proposal deviation are evaluated under the true prior μ, even when voters at its globally off-path ballot hold a different declared belief. Each result states whether it concerns all PBEs, a pure on-path proposal class, a constructed weak PBE, or a selection-free bound.

Players vote yes when indifferent. If a weak proposer is indifferent among payoff-maximizing proposals, the selected proposal minimizes H's expected payoff. No third tie-break is imposed.

For an R1 ballot information set $I = (h, s_i)$, let v_W denote the simultaneous weak-vote vector, let $\sigma_W(v_W \mid I)$ be the proposal-indexed distribution induced by the weak voting strategies, and let $A_R^Y(v_W)$ indicate implementation when H votes yes. The continuation history $h_2 = h_2(h, s_i, Y, v_W)$ includes the proposal, complete revealed vote vector, quota outcome, and opt-out status. The correct incentive constraint is

$$\sum_{v_W} \sigma_W(v_W \mid I) \left[A_R^Y(v_W) y + \{1 - A_R^Y(v_W)\} \beta C_{H,2}^R(\theta, h_2)\right] \geq o_\theta. \tag{6}$$

The right-hand side is always the immediate opt-out payoff. Only when $A_R^Y(v_W) = 1$ with probability one does this reduce to

$$y \geq o_\theta. \tag{7}$$

Appendix A states the corresponding weak-voter IC and records every voting history.

The main theorems use the regular interior domain

$$\mu \in (0,1), \qquad \beta \in (0,1), \qquad 0 < o_0 < o_1 < 1. \tag{8}$$

The boundaries $o_0 = 0$, $o_1 = 1$, $\beta = 1$, and degenerate priors are separated in Appendix C rather than imported by continuity.

Object	Scope
Terminal unanimity payoff	All-PBE payoff result in the terminal subgame.
Regular R1 unanimity	Global PBE existence and on-path payoff characterization under weak PBE and the declared completion beliefs.
Regular R1 majority	Pure on-path proposal characterization; tied mixtures inherit the stated payoff bounds.
Institutional comparison	Selection-free only on the derived common PBE-existence domain.
Boundaries and endpoints	Separate propositions or one-sided equilibrium-outcome limits; not continuations of the regular theorem by assumption.

Table 1: Scope of the clean baseline results.

5 Backward Induction

5.1 Terminal Round

Lemma 1 (Terminal unanimity)**.** *At posterior ν, define*

$$P = 1 - o_1, \qquad L(\nu) = (1-\nu)(1-o_0), \qquad S(\nu) = \max\{P, L(\nu)\}. \tag{9}$$

Every terminal unanimity PBE gives the weak coalition total payoff $S(\nu)$. The proposer offers $y = o_0$ when

$$\nu \le \nu_2^* = \frac{o_1 - o_0}{1 - o_0} \tag{10}$$

and $y = o_1$ above the cutoff. At equality, the proposal tie-break selects low-only. Each weak state's ex ante Round-2 value is $S(\nu)/m$.

Proposition 1 (Terminal majority)**.** *In terminal majority bargaining, all weak states approve. A weak proposer sets named weak offers to zero and retains the unit pie whether H is active or has* *already opted out. Therefore*

$$C^M_{W,2} = \frac{1}{m}, \qquad c = \frac{\beta}{m}, \qquad C^M_{H,2}(\theta) = o_\theta. \tag{11}$$

This result is selection-free, and H's outside payoff remains external to the weak pie.

The terminal screening result is the clean one-shot bridge. It does not imply that Round-1 acceptance follows a global cutoff: equation (6) still integrates simultaneous weak votes and the continuation after H-yes.

5.2 Round 1 Under Unanimity

For the regular domain, write

$$D_0 = 1 - o_0, \quad L = (1-\mu)D_0, \quad \delta = \frac{\beta(m-1)}{m}, \quad a = 1-\delta, \quad d = \frac{\beta P}{m}. \tag{12}$$

If pooling is on path, Bayes' rule prices a weak-voter deviation at $\beta S(\mu)/m$, and the maximal proposer payoff is

$$\pi_P = P - \delta S(\mu). \tag{13}$$

If low-only is on path, H-yes reveals the low type, so the maximal proposer payoff is

$$\pi_L = aL. \tag{14}$$

Globally off path, the proposer can guarantee two different security values:

$$G_L = (1-\mu)(D_0 - \delta P), \qquad G_P = aP. \tag{15}$$

The gap between the off-path low-only guarantee and its on-path payoff is strict:

$$G_L - \pi_L = (1-\mu)\delta(D_0 - P) > 0. \tag{16}$$

This discontinuity is why the old global pooling/low-only/delay reduction does not survive the clean game.

Lemma 2 (Off-path completion and guarantees)**.** *Suppose the regular domain holds and $\beta o_1 \geq o_0$. There is a weak-PBE completion that holds every globally off-path proposal to at most* $\max\{G_L, G_P\}$*. Conversely, the proposer can guarantee G_L with $y = o_0$ and $x_j = d$ for every non-proposer, and can guarantee G_P with $y = o_1$ and the same named offers. If $\beta o_1 < o_0$, the feasible proposal $y = o_1, x_j = 0$ has no sequentially rational ballot completion, so the full game has no PBE.*

Define

$$D_U = D_0 - \delta P > 0, \qquad \mu_E = 1 - \frac{aP}{D_U}. \tag{17}$$

Theorem 1 (Regular unanimity existence and payoff)**.** *On the regular interior domain, a global unanimity PBE exists if and only if*

$$\boxed{\beta o_1 \geq o_0 \quad \textit{and} \quad G_P > G_L,} \tag{18}$$

equivalently, $\beta o_1 \geq o_0$ and $\mu > \mu_E$. At equality $G_P = G_L$, the proposal tie-break selects a lower-H-payoff deviation whose on-path approval price jumps, so no fixed point exists.

Every regular unanimity PBE in the existence domain has the same on-path outcome:

$$y = o_1, \qquad T_W^U = P, \qquad V_W^U = \frac{P}{m}, \qquad E[u_H^U] = o_1. \tag{19}$$

Theorem 1 is a pooling result under the clean immediate opt-out contract. Low-only and rejection are not regular on-path equilibrium classes. Rejection can reappear when $\beta = 1$, and low-only can reappear when $o_0 = 0$; Appendix C gives the exact boundary statements. Nothing in the theorem asserts uniqueness of strategies or characterizes all PBEs under a stronger consistency concept.

5.3 Round 1 Under Majority

Let

$$k = q-1, \qquad c = \frac{\beta}{m}, \qquad E = 1-kc, \qquad B_M = (1-\mu)(1-o_0)+\mu c, \qquad F_M = \max\{E, B_M, P\}. \tag{20}$$

The three guarantees correspond to exclusion, a low-only offer, and pooling. For a non-proposing weak voter,

$$EU_j(Y) - EU_j(N) = (x_j - c)\,\Pr(j \text{ is pivotal}). \tag{21}$$

The exact security value is F_M, but implementability at that value depends on group size.

Proposition 2 (Regular majority correspondence)**.** *On the regular interior domain:*

1. If $N = 3$, the proposer payoff is exactly F_M. Exclusion is selected when E maximizes; low-only is selected when B_M maximizes; pooling requires $P > \max\{E, B_M\}$. A tie $E = B_M$ remains set-valued. 2. If $N = 4$, a majority PBE exists if and only if

$$E \geq B_M \quad \textit{or} \quad P > B_M. \tag{22}$$

Exclusion obtains when $E \geq \max\{B_M, P\}$, and pooling obtains when $P > \max\{B_M, E\}$. The boundary $B_M = P > E$ has no PBE. 3. If $N \geq 5$, a majority PBE always exists and the exact proposer-payoff set is

$$V^M_{\text{prop}} \in [F_M, 1]. \tag{23}$$

Every regular pure on-path proposal is exclusion, separating current passage, or pooling. A separating proposal $o_0 \leq y < o_1$ has

$$T^M_W = 1 - (1-\mu)y, \qquad E[u^M_H] = (1-\mu)y + \mu o_1. \tag{24}$$

Across all existing regular majority PBEs,

$$F_M \leq T^M_W \leq 1, \qquad \frac{F_M}{m} \leq V^M_W \leq \frac{1}{m}. \tag{25}$$

The historical No-Cheap-H condition also changes. Since

$$B_M - E = kc - o_0 + \mu(o_0 + c - 1), \tag{26}$$

the uniform sufficient condition is

$$o_0 \geq kc = (q-1)\frac{\beta}{m}. \tag{27}$$

It selects exclusion for $N = 3$ and guarantees an exclusion equilibrium for $N = 4$. For $N \geq 5$, it does not eliminate separating current passage. Majority no-screening is therefore not an all-PBE result.

6 Entry and Conditional Institutional Comparison

The comparison is restricted to the common regular PBE domain. Conditions (18) are required for unanimity. For $N = 4$, condition (22) is also required; for $N = 3$ and $N \geq 5$, majority existence is automatic once the regular domain holds. These are derived existence conditions rather than additional primitive assumptions.

Proposition 3 (Entry nesting)**.** *On the common PBE domain, every majority equilibrium satisfies*

$$V_W^M \geq \frac{F_M}{m} \geq \frac{P}{m} = V_W^U. \tag{28}$$

Consequently, collective formation sets satisfy

$$\mathcal{F}_U \subseteq \mathcal{F}_M \tag{29}$$

selection-free. Given a selected majority payoff $v_M = V_W^M$:

1. both institutions form if $\chi \leq P/m$; 2. only majority forms if $P/m < \chi \leq v_M$; 3. neither forms if $\chi > v_M$.

There is no region in which only unanimity forms.

Proposition 4 (Hegemon payoff bounds)**.** *Let*

$$\bar{o} = (1-\mu)o_0 + \mu o_1.$$

On the common PBE domain, every regular majority equilibrium satisfies

$$\bar{o} \le E[u_H^M] \le o_1 = E[u_H^U]. \tag{30}$$

Thus, conditional on both institutions forming, unanimity weakly benefits H, with equality under majority pooling. If only majority forms, majority weakly benefits H relative to nonformation under unanimity. If neither forms, H receives $\bar{o}$ under both rules.

Propositions 3 and 4 provide an institutional classification, not global dominance. They do not collapse the majority correspondence to a scalar payoff and do not compare rules at parameter vectors where one of the required PBEs does not exist.

7 Discussion

7.1 OPEC and pivotal participation

The OPEC interpretation is institutional rather than monetary. The package y can represent a quota share, production flexibility, an exception, or an enforcement arrangement that benefits a pivotal producer and reduces the residual institutional surplus available to other members. The private object o_θ is the producer's outside payoff from opting out: it can summarize opportunity costs, alternative production strategies, spare capacity, and other privately known advantages of nonparticipation.

The model does not claim that a literal OPEC meeting implements the exact two-round protocol. It isolates a mechanism: when every agreement needs the pivotal producer's approval, weaker members must insure against a private participation constraint. In the regular existence region, this pressure produces pooling at the high outside payoff even though the hegemon never proposes. Under majority, weak coalitions can exclude H, but the equilibrium correspondence can also include separating current passage when the group is large enough. The empirical implication is therefore not that majority mechanically removes information rents; it is that institutional rules change whether and how a private participation constraint enters the coalition problem.

Observable implications and neighboring mechanisms

Four implications follow directly from the clean results. First, terminal unanimity switches from low-only to pooling as the posterior probability of the high outside-payoff type rises. Second, regular Round-1 unanimity is more demanding: equilibrium may fail to exist, but every existing on-path outcome pools at o_1. Third, majority outcomes are group-size dependent and set-valued; for $N \geq 5$, exclusion and separating current passage can both survive. Fourth, entry nesting is selection-free only on the common PBE-existence domain.

These implications differ from agenda power and endogenous institutional choice. A positive recognition probability for H would give the informed actor proposal power and introduce a signaling subgame. Allowing H to choose the voting rule after observing its type could make the rule itself a signal. Neither channel is part of the baseline.

7.2 Scope

The baseline fixes $\pi_H = 0$, $b_\theta = 0$, a unit weak pie, binary private information, simultaneous ballots, and exogenous voting rules. A no vote by H is an immediate opt-out; it is not a choice of βC_θ, a hybrid $\max\{o_\theta, \beta C_\theta\}$, or a delayed-continuation threshold. The package space is always feasible and does not revive the archived feasibility/C-B-R architecture.

The weak-vote-passive assessment is also a substantive scope condition. It is informationally natural because weak states do not observe θ, but weak PBE leaves globally off-path beliefs freer than sequential equilibrium. Theorem 1 states exactly where this freedom is used, and no claim of uniqueness over all PBEs is made.

Boundary parameters matter. Low-only and rejection can reappear at $o_0 = 0$ or $\beta = 1$, and majority needs separate boundary bounds. These cases are recorded in Appendix C rather than treated as if they belonged to the regular interior theorem. Continuous types, heterogeneous weak states, $\pi_H > 0$, endogenous rule choice, and sequential roll-call voting require new derivations.

8 Conclusion

Consensus is often described only as a constraint on power. The clean immediate-opt-out model shows another possibility. When a powerful actor is privately informed and pivotal, its approval becomes an informational constraint on weaker proposers. This creates informational power even when the hegemon has no agenda power.

The result is conditional. Regular unanimity equilibrium exists only on a derived domain and pools whenever it exists. Majority equilibrium is generally a correspondence rather than a unique no-screening benchmark. On their common existence domain, weak-state formation under unanimity is nested in formation under majority, while unanimity weakly raises the hegemon's payoff whenever both institutions form. The institutional ranking therefore follows from existence, formation, and equilibrium selection—not from a blanket claim that one voting rule dominates.

The broader implication is that formal equality and asymmetric power can coexist through information. Consensus can make a privately known outside payoff politically valuable by turning a veto into a participation constraint. That mechanism is distinct from both proposal power and constitutional choice, and it survives precisely because the baseline keeps those channels separate.

Appendix A: Protocol and Histories

A.1 Public histories, beliefs, and voting incentives

Let h denote the public history before a proposal and let $I = (h, s_i)$ denote a ballot information set. Once all simultaneous votes are revealed, the public history is

$$h_2 = (h, s_i, a_H, v_W, \text{quota outcome, opt-out status}). \tag{A1}$$

Continuation strategies can therefore condition on the proposal and complete vote vector, not only on a posterior. Whenever Bayes' denominator is positive,

$$\nu(h_2) = \frac{\mu(h)\,\Pr(h_2 \mid \theta = 1, h)}{\mu(h)\,\Pr(h_2 \mid \theta = 1, h) + \{1 - \mu(h)\}\,\Pr(h_2 \mid \theta = 0, h)}. \tag{A2}$$

The posterior $\nu(h_2)$ and the continuation payoff $C^R_{H,2}(\theta, h_2)$ are distinct objects.

At an R2 information set I_2, let $p_R(I_2)$ be the probability that H-yes implements the current proposal, given simultaneous weak strategies. Then

$$EU^R_H(Y \mid \theta, I_2) - EU^R_H(N \mid \theta, I_2) = \beta p_R(I_2)(y - o_\theta). \tag{A3}$$

If $p_R(I_2) > 0$, the sign is that of $y - o_\theta$. If $p_R(I_2) = 0$, H is indifferent for every y and votes yes by the tie rule. This terminal identity does not replace the Round-1 expected IC in (6).

For a non-proposing weak voter j, let $u_j(a_j, v_{-j}; I)$ denote the payoff determined by the history table. Approval requires

$$\sum_{v_{-j}} \Pr(v_{-j} \mid I) u_j(Y, v_{-j}; I) \geq \sum_{v_{-j}} \Pr(v_{-j} \mid I) u_j(N, v_{-j}; I). \tag{A4}$$

The expectation includes H's simultaneous type-contingent vote. Named offers are paid whenever an agreement passes, even if the named recipient voted no.

A.2 Exhaustive history contract

Table panels 2–5 are generated from the versioned source `tables/clean_optout_gate0_histories_piH0.tsv`. Together they cover the twenty-one history classes used by the proofs.

Table 2: Gate 0 histories, panel A: timing, electorate, quota, and votes.

history_id	round	rule	active players	quota	quota reached?	H vote
G01	R1	U	H plus m weak	N	yes: z=m	yes
G02	R1	U	H plus m weak	N	no: z<m	yes
G03	R1	U	H plus m weak	N	no: z=m	no
G04	R1	U	H plus m weak	N	no: z<m	no
G05	R1	M	H plus m weak	q	yes: z>=q-1	yes
G06	R1	M	H plus m weak	q	no: z<=q-2	yes
G07	R1	M	H plus m weak	q	yes: z>=q	no
G08	R1	M	H plus m weak	q	no: z=q-1	no
G09	R1	M	H plus m weak	q	no: z<=q-2	no
G10	R2	U	H plus m weak	N	yes: z=m	yes
G11	R2	U	H plus m weak	N	no: z<m	yes
G12	R2	U	H plus m weak	N	no: z=m	no
G13	R2	U	H plus m weak	N	no: z<m	no
G14	R2	M	H plus m weak	q	yes: z>=q-1	yes
G15	R2	M	H plus m weak	q	no: z<=q-2	yes
G16	R2	M	H plus m weak	q	yes: z>=q	no
G17	R2	M	H plus m weak	q	no: z=q-1	no
G18	R2	M	H plus m weak	q	no: z<=q-2	no
G19	R2	M	m weak; H already opted out	original q	yes: z>=q	not applicable
G20	R2	M	m weak; H already opted out	original q	no: z<q	not applicable
G21	R2	U	m weak; H already opted out	N	structurally impossible	not applicable

Table 3: Gate 0 histories, panel B: implementation and the disposition of the participation package.

history_id	participation offer to H?	failure cause	H included?	H remains?	y implemented?	destination of y
G01	yes	none	yes	no continuation	yes	paid to H
G02	yes	weak-vote shortfall	no	yes	no	not implemented
G03	yes	H-no is decisive	no	no	no	not implemented
G04	yes	joint H-no and weak rejection	no	no	no	not implemented
G05	yes	none	yes	no continuation	yes	paid to H
G06	yes	weak-vote shortfall	no	yes	no	not implemented
G07	yes	none; weak votes reach q	no	no	no	reabsorbed by proposer
G08	yes	H-no is decisive for failure	no	no	no	not implemented
G09	yes	joint H-no and weak shortfall	no	no	no	not implemented
G10	yes	none	yes	no continuation	yes	paid to H
G11	yes	weak-vote shortfall	no	no	no	not implemented
G12	yes	H-no is decisive	no	no	no	not implemented
G13	yes	joint H-no and weak rejection	no	no	no	not implemented
G14	yes	none	yes	no continuation	yes	paid to H
G15	yes	weak-vote shortfall	no	no	no	not implemented
G16	yes	none; weak votes reach q	no	no	no	reabsorbed by proposer
G17	yes	H-no is decisive for failure	no	no	no	not implemented
G18	yes	joint H-no and weak shortfall	no	no	no	not implemented
G19	no; y fixed at 0	none	no	no	no; fixed at 0	not available
G20	no; y fixed at 0	weak-vote shortfall	no	no	no; fixed at 0	not available
G21	no; y fixed at 0	H absent, so unanimity quota cannot be met	no	no	no; fixed at 0	not available

Table 4: Gate 0 histories, panel C: payoffs and time units.

history_id	H payoff	weak payoffs	discount factor
G01	y	x_j; proposer gets 1-y-sum(x_j)	1
G02	beta*C_H2^U(theta,h2)	beta times R2-U continuation payoffs	beta
G03	o_theta	0	1
G04	o_theta	0	1
G05	y	x_j; proposer gets 1-y-sum(x_j)	1
G06	beta*C_H2^M(theta,h2)	beta times R2-M continuation payoffs with H active	beta
G07	o_theta	x_j; proposer gets 1-sum(x_j)	1
G08	o_theta	beta times R2 weak-only continuation payoffs	beta for weak continuation; H paid at R1
G09	o_theta	beta times R2 weak-only continuation payoffs	beta for weak continuation; H paid at R1
G10	beta*y	beta times x_j; proposer gets beta*[1-y-sum(x_j)]	beta
G11	beta*o_theta	0	beta
G12	beta*o_theta	0	beta
G13	beta*o_theta	0	beta
G14	beta*y	beta times x_j; proposer gets beta*[1-y-sum(x_j)]	beta
G15	beta*o_theta	0	beta
G16	beta*o_theta	beta times x_j; proposer gets beta*[1-sum(x_j)]	beta
G17	beta*o_theta	0	beta
G18	beta*o_theta	0	beta
G19	0 additional	beta times x_j; proposer gets beta*[1-sum(x_j)]	beta
G20	0 additional	0	beta
G21	0 additional	0	not applicable

Table 5: Gate 0 histories, panel D: beliefs and continuation subgames.

history_id	relevant posterior	terminal or continuation	next subgame
G01	irrelevant after agreement	terminal	none
G02	nu(h2) from proposal and full vote vector; continuation may depend on full h2	continuation	R2 U with H active
G03	irrelevant after opt-out	terminal	none
G04	irrelevant after opt-out	terminal	none
G05	irrelevant after agreement	terminal	none
G06	nu(h2) from proposal and full vote vector; continuation may depend on full h2	continuation	R2 M with H active
G07	irrelevant after opt-out	terminal	none
G08	irrelevant after opt-out	continuation	R2 M weak-only
G09	irrelevant after opt-out	continuation	R2 M weak-only
G10	irrelevant after terminal agreement	terminal	none
G11	irrelevant at terminal date	terminal	none
G12	irrelevant at terminal date	terminal	none
G13	irrelevant at terminal date	terminal	none
G14	irrelevant after terminal agreement	terminal	none
G15	irrelevant at terminal date	terminal	none
G16	irrelevant at terminal date	terminal	none
G17	irrelevant at terminal date	terminal	none
G18	irrelevant at terminal date	terminal	none
G19	theta is payoff-irrelevant	terminal	none
G20	theta is payoff-irrelevant	terminal	none
G21	theta is payoff-irrelevant	unreachable; branch ended at opt-out	none

The R2-unanimity row with H already absent documents a structurally unreachable class. Once H opts out, the original unanimity quota cannot be met and that branch has already ended.

Appendix B: Proofs of Regular Results

B.1 Terminal rounds

Proof of Lemma 1. In terminal unanimity bargaining, every non-proposing weak voter approves: a no vote either leaves failure unchanged or changes failure into an agreement paying a nonnegative named offer, and ties go to yes. Hence H is pivotal, and equation (A3) gives the cutoff $y \geq o_\theta$.

A package below o_0 fails for both types. Within $[o_0, o_1)$, $y = o_0$ maximizes weak surplus and is accepted only by the low type, giving $L(\nu)$. At or above o_1, $y = o_1$ maximizes weak surplus and pools, giving P. Comparing these values yields (10). At equality the weak proposer is indifferent and the proposal tie-break selects the smaller transfer to H. Uniform recognition divides the total weak value by m. □

Proof of Proposition 1. Terminal failure gives weak states zero, so weak approval is weakly dominant. All weak states approve. Because $q \leq m$, a weak proposer can set every named offer to zero, pass with weak votes alone, and retain the unit pie. This construction is valid whether H is active or has opted out. If H is active and votes no, it receives o_θ externally; if it votes yes, the proposal can set $y = 0$. Uniform recognition gives the representative weak payoff $1/m$, and discounting gives $c = \beta/m$ in Round-1 units. □

B.2 Unanimity completion and existence

Proof of Lemma 2. First suppose $\beta o_1 \geq o_0$. For each globally off-path proposal in the upper-bound construction, retain a ballot belief assigning probability $1 - \mu \in (0, 1)$ to the low type. After a history with low-type H-yes and exactly one weak no, declare a continuation posterior $\nu = 1 > \nu_2^*$, so terminal R2 selects pooling. Because the proposal itself has probability zero, weak PBE does not link these two beliefs by Bayes' rule. The low type then obtains βo_1 after weak-caused failure and is willing to vote yes; the high type opts out.

Complete globally off-path proposals as follows.

1. If $y < o_0$, prescribe all weak voters yes and both types of H no. The proposer obtains zero.
2. If $o_0 \leq y < o_1$ and every named weak offer is at least $d = \beta P/m$, prescribe all weak voters yes, low-type H yes, and high-type H no. The best such proposal gives at most G_L.
3. If $y \geq o_1$ and every named weak offer is at least d, prescribe all weak voters and both types of H yes. The best such proposal gives at most G_P.
4. If some $x_j < d$, designate one underpaid weak rejector, prescribe low-type H yes and high-type H no, and use the pooling continuation. The rejector strictly prefers continuation, and the proposer's payoff is at most $(1-\mu)\beta P/m < G_L$.

Two or more weak no votes cannot occur because each rejector would be nonpivotal and would have to vote yes by the tie rule. A nondegenerate ballot mixture cannot enlarge the set because indifference also selects yes.

For the reverse direction, consider $y = o_0$ and $x_j = d$ for every non-proposer. In any sequentially rational completion, all weak voters vote yes. Two no votes violate the tie rule. With exactly one weak no, that voter can strictly prefer rejection only if terminal continuation exceeds d, which requires R2 low-only. But low-only pays low-type H $\beta o_0 < o_0$ after H-yes, so the low type opts out and makes the weak rejector nonpivotal. With all weak voters yes, low-type H votes yes, high-type H votes no, and the proposer receives exactly G_L.

The same argument applies to $y = o_1$ with $x_j = d$. A lone weak rejector again needs low-only continuation. Both types then prefer immediate opt-out to the discounted terminal outside payoff after H-yes. The rejector becomes nonpivotal, all weak voters vote yes, both types of H approve, and the proposer obtains G_P. Thus G_L and G_P are genuine guarantees and the construction gives the matching upper bound.

If $\beta o_1 < o_0$, the feasible proposal $y = o_1, x_j = 0$ has no sequentially rational ballot completion. All weak yes makes each weak voter prefer a positive R2 continuation; two weak no votes violate the tie rule; and with exactly one weak no both types of H opt out. Hence no PBE exists on that open subdomain. □

Proof of Theorem 1. Lemma 2 shows that the proposer can force both G_L and G_P, and that every proposal can be completed at no more than their maximum. Yet an on-path low-only proposal must pay each non-proposer $\beta D_0/m > d$, because H-yes reveals the low type. Therefore

$$G_L - \pi_L = (1-\mu)\delta(D_0 - P) > 0.$$

Low-only cannot be a regular PBE: off path the proposer shaves named weak offers to d, but once that proposal becomes on path Bayes' rule restores the higher approval price.

Pooling attains its security value only when $P > L$, in which case $\pi_P = aP = G_P$. Direct algebra gives

$$\mu_E - \nu_2^* = \frac{\delta P(D_0 - P)}{D_0 D_U} > 0, \tag{B1}$$

so $G_P > G_L$ implies $P > L$. At $G_P = G_L$, the intermediate deviation ties the proposer payoff and pays less to H, so the proposal tie-break selects it; once on path, its approval price jumps and the proposed fixed point fails. Under the strict inequality, pooling strictly dominates every completed deviation and Lemma 2 constructs a global PBE. The pooling package gives the payoff vector in (19). □

B.3 Majority security and group-size characterization

Proof of Proposition 2. Let $X = \sum_{j \neq i} x_j$, and let r count non-proposers offered at least c. If at least $q-1$ weak states vote yes, H-yes implements and type θ approves iff $y \geq o_\theta$. If fewer than $q-1$ weak states vote yes, H-yes still fails and pays only $\beta o_\theta < o_\theta$, so both types opt out. Outside the all-weak-yes profile, the only regular pure ballot size is therefore $q-1$.

The upper-bound completion has three cases. If $y < o_0$ and $r \leq k-1$, a $q-1$-weak-yes rejection completion pays the proposer c; if $r \geq k$, weak-only passage gives at most $1 - X \leq E$. If $o_0 \leq y < o_1$, declare the globally off-path ballot belief $\nu = 1$. With $r \leq k-1$, choose $q-1$ weak yes votes; the true low state passes, the true high state continues, and the proposer obtains at most B_M. With $r \geq k$, weak-only passage is forced and yields less than E. If $y \geq o_1$, all weak voters are nonpivotal when they all vote yes, and the proposer receives at most P.

Conversely, paying c to k weak partners guarantees E; $y = o_0, x = 0$ guarantees at least B_M; and $y = o_1, x = 0$ guarantees P. Hence F_M is the exact security value.

For $N = 3$, all-weak-yes is a minimum winning coalition. Each of the three security constructions is implementable when it maximizes, so proposer payoff is F_M. Exclusion and low-only remain set-valued when $E = B_M$ because both give H its outside payoff; pooling survives only under the stated strict inequality. Their respective weak totals are

$$T_W^E = 1, \qquad T_W^B = (1-\mu)(1-o_0) + \mu\beta, \qquad T_W^P = P. \tag{B2}$$

For $N = 4$, the low-only proposal guarantees B_M off path but has no Bayes-consistent pure on-path ballot. Only exclusion and pooling can attain the security value. This gives condition (22) and the two selection regions. When $B_M = P > E$, the unattainable deviation ties pooling in proposer payoff and gives H less, so the proposal tie-break destroys the pooling fixed point.

For $N \geq 5$, all-weak-yes is oversized because $q \leq m - 1$. For any $v \in [F_M, 1]$, select an exclusion proposal with $y = 0$ and total named weak offers $X = 1 - v$. Every non-proposer is nonpivotal at all-weak-yes and votes yes by the tie rule; the weak-only proposal passes and gives the proposer v. Complete every deviation with the security construction at no more than F_M. This proves sufficiency throughout the interval, including its floor.

Any regular pure on-path proposal is then one of three classes. Exclusion has $y < o_0$, weak total one, and pays H its outside payoff. Separating current passage has $o_0 \leq y < o_1$: low-type H votes yes, high-type H votes no, and the oversized weak coalition passes in both states. With named offers X, proposer payoff and total weak payoff are

$$v = 1 - X - (1-\mu)y, \qquad T_W^M = 1 - (1-\mu)y.$$

The canonical $y = o_0, X = 0$ class exists iff $1 - (1-\mu)o_0 \geq E$. Pooling uses $y = o_1, X = 0$ and exists iff $P > \max\{E, B_M\}$. These classes and their mixtures imply (25). □

Equation (26) is affine in μ. Its endpoint gaps are $kc - o_0$ and $qc - 1 \leq 0$, giving condition (27). The group-size implications follow directly from Proposition 2.

B.4 Institutional comparison

Proof of Proposition 3. On the common PBE domain, Theorem 1 gives $V_W^U = P/m$. Proposition 2 gives $V_W^M \geq F_M/m$, and $F_M \geq P$ by definition. This proves (28), formation-set inclusion, and the three cost regions. □

Proof of Proposition 4. Exclusion pays H exactly $\bar{o}$. Separating current passage pays

$$(1-\mu)y + \mu o_1 \in [\bar{o}, o_1)$$

because $y \in [o_0, o_1)$. Pooling pays o_1. Convex combinations of payoff-tied majority classes preserve these bounds. Unanimity pays o_1 on the common domain by Theorem 1. This proves (30) and the conditional classification. □

Appendix C: Boundaries and Endpoint Limits

C.1 Zero low outside payoff

Let $\beta < 1$, $o_0 = 0$, $0 < o_1 \leq 1$, and $\mu \in (0,1)$. Retain

$$P = 1 - o_1, \qquad L = 1 - \mu, \qquad \delta = \frac{\beta(m-1)}{m}, \qquad a = 1 - \delta,$$

and define

$$K_0 = \max\{aL, P - \delta\}. \tag{C1}$$

The second term is inactive when negative.

Proposition 5 (Unanimity when $o_0 = 0$)**.** *A global PBE always exists. If $L \geq P$, low-only is the selected accepted class and gives $T_W^U = L$. If $P > L$, canonical pooling exists and gives $T_W^U = P$; low-only also exists if and only if*

$$P - \delta \leq aL. \tag{C2}$$

At equality the proposal tie-break favors low-only. If $P > L$, an overpaid pooling class with

$o_1 < y \le \bar{y}$ and $T = 1 - y$ exists exactly when

$$T \in [1 - \bar{y}, P), \qquad T - \delta P > K_0. \tag{C3}$$

Rejection is never on path on this boundary.

Proof. Terminal low-only gives each weak state $1/m$, so after a globally off-path proposal it can make the approval price β/m. On-path low-only gives the proposer aL. A pooling proposal that pays β/m to every non-proposer forces approval and gives $P-\delta$. These two constructions give the exact security value K_0. Deliberate R1 rejection gives its original proposer only $(1-\mu)\beta/m < aL$.

If $L \ge P$, low-only attains the security value. If $P > L$, canonical pooling gives $aP > K_0$, while low-only remains optimal under another off-path completion exactly under (C2). At $L = P$, the tie-break selects low-only because it pays less to H. An overpaid pooling proposal gives proposer payoff $T - \delta P$ and survives precisely when it strictly exceeds both security components. □

Low-only gives H expected payoff μo_1, canonical pooling gives o_1, and overpaid pooling gives $1 - T$.

C.2 No terminal weak surplus

Proposition 6 (Unanimity when $o_1 = 1$)**.** *If $\beta < 1$, $o_0 > 0$, $o_1 = 1$, and $\mu \in (0, 1)$, no unanimity PBE exists.*

Proof. Here $P = 0$. The proposal $y = o_0, x_j = 0$ forces weak approval. Two weak no votes make each rejector nonpivotal. With exactly one weak no, a positive continuation requires terminal low-only, but low-type H then receives only $\beta o_0 < o_0$ after voting yes and opts out, again making the rejector nonpivotal. With zero continuation, the weak yes tie rule applies. Low-type H therefore approves, high-type H opts out, and the deviation guarantees the proposer $L = (1 - \mu)(1 - o_0)$. On-path low-only yields only $aL < L$, pooling leaves zero weak surplus, and rejection gives the original proposer at most $\beta L/m < L$. No on-path class attains the guarantee. □

C.3 No discounting

Let $\beta = 1$, $0 \leq o_0 < o_1 \leq 1$, and $\mu \in (0, 1)$. Define

$$K_1 = \max\left\{\frac{L}{m}, P - \frac{(m-1)(1-o_0)}{m}\right\}. \tag{C4}$$

Proposition 7 (Unanimity when $\beta = 1$)**.** *Rejection or continuation is admissible and gives the R1 proposer $S(\mu)/m$. If $L \geq P$, low-only and rejection survive and pooling is selected against. If $P > L$, pooling and rejection survive; low-only also survives if and only if*

$$P - \frac{(m-1)(1-o_0)}{m} \leq \frac{L}{m}. \tag{C5}$$

For $P > L$, an overpaid pooling class with $o_1 < y \leq \bar{y}$ and $T = 1 - y$ exists exactly when

$$T - \frac{m-1}{m}P > K_1. \tag{C6}$$

Proof. With no discounting, after weak-caused failure low-type H is indifferent between terminal low-only and immediate opt-out, and high-type H is indifferent between terminal and immediate opt-out. The yes tie rule sustains the punitive continuation. Low-only therefore gives and can guarantee L/m, while a pooling proposal that pays $(1 - o_0)/m$ to every non-proposer guarantees the second component of K_1 when feasible. A deliberately rejected proposal gives $S(\mu)/m$. The comparisons in (C5) and (C6) then give the stated classes; equality is selected against the proposal that pays H more. Mixtures over payoff- and H-payoff-equivalent classes can survive. □

This is the only baseline locus on which the historical rejection or delay label survives on path. The intersections $o_0 = 0$ and $o_1 = 1$ are included in Proposition 7.

C.4 Majority boundary bounds

At $\beta = 1$, or at $o_0 = 0$ with $\beta < 1$, H-yes can be optimal even when only $q - 2$ weak states vote yes and the current ballot still fails. Rejection classes can therefore reappear. For $N \geq 4$, the corresponding algebraic floors are

$$F_{M,1} = \max\{1 - kc,\ (1-\mu)[1 - o_0 - (k-1)c] + \mu c,\ 1 - o_1 - (k-1)c\}, \tag{C7}$$

and

$$F_{M,0} = \max\{1 - kc,\ (1-\mu)[1 - (k-1)c] + \mu c,\ 1 - o_1 - (k-1)c\}. \tag{C8}$$

These are boundary bounds, not complete equilibrium characterizations and not substitutes for Proposition 2. No institutional comparison in the paper extrapolates the regular theorem to these loci.

C.5 Degenerate priors as one-sided limits

Literal games with $\mu = 0$ or $\mu = 1$ require an extra convention about zero-probability types. The paper instead reports one-sided limits of regular equilibrium outcomes. Let

$$z = (v_{\text{prop}}, T_W, V_W, E[u_H]), \qquad V_W = T_W/m,$$

and let $\mathcal{E}_R(\mu)$ be the regular on-path PBE-outcome correspondence. For $e \in \{0, 1\}$, define

$$\mathcal{L}_e^R = \left\{\lim_{n\to\infty} z_n : \mu_n \in (0,1),\ \mu_n \to e,\ z_n \in \mathcal{E}_R(\mu_n)\right\}. \tag{C9}$$

An empty correspondence means that no sequence of regular PBEs approaches the endpoint, not that a separately specified degenerate game lacks a PBE.

For unanimity,

$$\mathcal{L}_0^U = \emptyset, \qquad \mathcal{L}_1^U = \begin{cases} \{(aP, P, P/m, o_1)\}, & \beta o_1 \geq o_0, \\ \emptyset, & \beta o_1 < o_0. \end{cases} \tag{C10}$$

For majority, write

$$B_0^+ = D_0, \qquad B_1^- = c, \qquad F_0^+ = \max\{E, D_0\}, \qquad F_1^- = \max\{E, P\}. \tag{C11}$$

Group	$\mu \to 0^+$	$\mu \to 1^-$
$N = 3$	Low-only $(D_0, D_0, D_0/m, o_0)$ if $D_0 > E$; otherwise exclusion $(E, 1, 1/m, o_0)$.	Pooling $(P, P, P/m, o_1)$ if $P > E$; otherwise exclusion $(E, 1, 1/m, o_1)$.
$N = 4$	Empty if $D_0 > E$; otherwise exclusion $(E, 1, 1/m, o_0)$.	Pooling $(P, P, P/m, o_1)$ if $P > E$; otherwise exclusion $(E, 1, 1/m, o_1)$.

Table 6: One-sided majority equilibrium-outcome limits for $N = 3$ and $N = 4$.

At $D_0 = E$, the lower lateral tie belongs to exclusion; at $P = E$, the upper lateral tie also belongs to exclusion. For $N \geq 5$,

$$\mathcal{L}_0^M = \begin{cases} \{(v, 1, 1/m, o_0) : v \in [E, 1]\}, & D_0 < E, \\ \{(v, 1, 1/m, o_0) : v \in [D_0, 1]\} \cup \{(D_0, t, t/m, o_0) : t \in [D_0, 1]\}, & D_0 \geq E, \end{cases} \tag{C12}$$

and

$$\mathcal{L}_1^M = \{(v, 1, 1/m, o_1) : v \in [F_1^-, 1]\} \cup \begin{cases} \{(P, P, P/m, o_1)\}, & P > E, \\ \emptyset, & P \leq E. \end{cases} \tag{C13}$$

Derivation. The affine guarantee satisfies $B_M(\mu) \to D_0$ at 0^+ and $B_M(\mu) \to c$ at 1^-. Substitution into the exact small-group characterizations gives Table 6. For $N \geq 5$, exclusion supplies the proposer intervals. Near zero, a separating class has weak total $1 - (1 - \mu)y$; attaining the floor forces $y \to o_0$ when $D_0 \geq E$. Near one, exclusion and separation converge to weak total one and H payoff o_1, while canonical pooling adds a distinct limit exactly when $P > E$. These exhaust the regular classes. □

There is no common lower-endpoint comparison because $\mathcal{L}_0^U$ is empty. At the upper endpoint, a common limit exists iff $\beta o_1 \geq o_0$; entry nesting carries to every common selection, and all corresponding H-payoff limits equal o_1.

C.6 One-shot bridge

Suppressing Round 2 leaves exactly the terminal subgame in Lemma 1. The weak proposer compares $(1-\mu)(1-o_0)$ with $1-o_1$, so the low-only and pooling regions are separated by ν_2^*. This is an interpretation of the proved terminal subgame, not a new complete-information benchmark and not a second baseline game.

Appendix D: Notation

Table 7: Notation for the clean immediate-opt-out $\pi_H = 0$ baseline.

Symbol	Type	Meaning	First use
N	Primitive	Total number of states	Definition 1
$m = N-1$	Derived	Number of weak states	Definition 1
q	Rule parameter	Original strict-majority quota	Equation (4)
$k = q-1$	Derived	Additional yes votes required by a weak proposer under majority	Equation (20)
θ	Type	Persistent hegemon type, low 0 or high 1	Definition 1
μ	Belief	Prior probability of the high type	Definition 1
o_θ	Primitive payoff	Immediate outside payoff after H-no; direct cutoff under certain implementation	Definition 1
y	Contract component	Participation package paid to H only when included	Definition 1
$\bar{y}$	Primitive bound	Maximum participation package	Equation (1)
x_j	Contract component	Named weak-state offer	Definition 1
β	Primitive	Discount factor for Round-2 payoffs in Round-1 units	Definition 1
π_H	Recognition parameter	Hegemon recognition probability, fixed at zero in the baseline	Introduction
χ	Entry cost	External sunk cost per weak state	Equation (5)
P	Derived value	Terminal pooling weak total, $1-o_1$	Equation (9)
$L(\nu)$	Derived value	Terminal low-only weak total	Equation (9)
$S(\nu)$	Derived value	Terminal unanimity weak security value	Equation (9)
D_0	Derived value	Low-type weak residual, $1-o_0$	Equation (12)
δ	Derived value	Aggregate discounted approval weight, $\beta(m-1)/m$	Equation (12)
a	Derived value	$1-\delta$	Equation (12)
G_L, G_P	Security values	Exact unanimity low-only and pooling guarantees	Equation (15)
μ_E	Existence cutoff	Regular unanimity existence threshold	Equation (17)
c	Continuation value	Weak voter's discounted majority continuation, β/m	Equation (11)
E	Security value	Majority exclusion guarantee, $1-kc$	Equation (20)
B_M	Security value	Majority low-only guarantee	Equation (20)
F_M	Security value	Exact majority proposer security value	Equation (20)
$\bar{o}$	Derived payoff	Prior expected outside payoff of H	Proposition 4
$\mathcal{F}_R$	Formation set	Beliefs or selections for which rule R forms at cost χ	Proposition 3
K_0, K_1	Boundary values	Unanimity security values at $o_0 = 0$ and $\beta = 1$	Appendix C

Symbol	Type	Meaning	First use
$\mathcal{L}_e^R$	Limit correspondence	One-sided cluster set of regular equilibrium outcomes	Equation (C9)

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